

LESSON 113 (0.5 GROUND and 1.5 DUAL)

Lesson Objective

During this lesson, the applicant will be introduced to maneuvering during slow flight. The applicant will learn the signs of an incipient and full stall as well as the differences between them. The applicant will also learn the skills and risk management associated with power-off and power-on stalls and stall prevention. The applicant will continue familiarizing themselves with radio communication procedures with instructor guidance

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed.
 - Ask the students if they have any questions on the procedures from the maneuver's manual.
 - Normal and crosswind takeoff.
 - Maneuvering during slow flight.
 - Power-off stalls.
 - Power-on stalls.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
- Aeronautical Information Manual.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
- Warrior Maneuvers Manual.

Instructor Guidance

- This lesson introduces some of the most safety-critical maneuvers in the entire course — slow flight, stalls, and the cross-controlled stall demonstration. Set the tone during the pre-brief by framing these maneuvers as essential tools for aeronautical decision-making and accident prevention, not just checkride items.
- Before departing for the practice area, ensure the student has a solid conceptual understanding of what a stall is and how it differs from an engine stall. Students who arrive without having read the assigned material will struggle significantly in the air — if preparation is clearly lacking, address it before the flight.
- Always use the telling and doing technique for every new maneuver introduced in this lesson. Demonstrate first with a full narration, then have the student attempt it. Do not hand over the controls on a stall entry without a proper demonstration.
- For slow flight, emphasize the pre-maneuver check and clearing turns before every entry. These habits must become automatic — reinforce them each and every time without

exception.

- When demonstrating the difference between incipient and full stall recovery, have the student note the altitude loss for each. The visual impact of seeing a larger altitude loss from a full stall versus an incipient recovery is one of the most effective teaching tools available for building stall awareness.
- For stall recovery, stress that the priority is always to reduce the angle of attack first — pitch forward before applying power. Students who instinctively jam in full power without reducing pitch risk a secondary stall. Demonstrate the consequence clearly.

Lesson Content (Ground)

Normal and Crosswind Takeoff

- Taxi into takeoff position when cleared for takeoff by ATC and verify magnetic heading matches assigned takeoff runway heading.
- Smoothly and continuously apply takeoff power.
- Check static RPM (approximately 2330 RPM) and verify airspeed rising.
- Use rudder as necessary to maintain directional control and ailerons (into the wind) to aid with crosswind correction.
- When the flight controls become effective, gradually apply back stabilator pressure to attain a lift-off pitch attitude. Ensure that ailerons are neutral prior to rotating.
- Note the position of the airplane's nose in relation to the horizon and adjust as necessary to attain a VY climb pitch attitude.
- At 1,000 feet AGL, lower the nose, approximately an inch, to attain a cruise climb pitch attitude and complete the climb check.

Climbing, Descending and Turning Techniques

- Review the lesson plan for lessons 108 and 109 as needed.

Stall and Spin Awareness

- Define what an aircraft upset is:
 - "Unusual attitude" or loss of "positive control".
 - Use "Chapter 5 - Maintaining Aircraft Control..." in the AFH.
- Define what a stall is:
 - Differentiate aerodynamic stalls from engine stalls. (Overuse of abstraction).
 - Rapid decrease in lift caused by the separation of airflow from the wings surface when exceeding the critical AoA.
 - Wing does not stop producing lift – it cannot generate adequate lift to sustain level flight.
 - Can occur at any pitch, any attitude or airspeed.
- Describe in general terms how a stall might occur, and how to correct for it.
- Define what a spin is.
- Describe in general terms what a spin is, and the common phase of flight a spin might occur.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Student should be able to perform the "passenger briefing" and the "engine fire during start brief".

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student understands how to properly use COMS and NAV units.
 - Points of Emphasis:
 - Equipment ON/OFF
 - Database Currency
 - Volume
 - Squelch
 - Standby/Active Frequencies
 - Switch between navigation sources
 - Other useful functions such as: NRST page and Direct-To

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Radio Communication Procedures and ATC Phraseology

- Students should understand and demonstrate:

- Proper radio setup and frequency management
- Initial call-up format (who you are calling, who you are, where you are, what you want)
- Accurate readback of clearances and instructions (runway assignments, hold short instructions, altitudes, headings, squawk codes)
- Use of standard terminology (e.g., "affirmative," "negative," "standby," "unable")
- Avoid nonstandard or casual language
- Maintaining situational awareness while communicating
- The student should be able to conduct radio calls in the practice area.
- The instructor should assist as needed.

Normal and Crosswind Takeoff

- Have the student vocalize the runway line-up checklist and verify all line items.
- Ensure the student establishes full power with one hand on the throttle and one hand on the yoke.
- Verify the student is inputting proper crosswind correction.
 - Ensure the student points out the windsock and corrects accordingly.
- Once off the ground, TRIM for desired airspeed!

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Climbing Turns

- Situational awareness and proper clearing procedures are critical.
- As angle of bank increases the vertical component of lift is reduced, thus causing the nose to drop.
- To compensate for this, as a turn is made, an increase in back pressure will be required to maintain the climb.
- TRIM.

Descending Turns

- Situational awareness and proper clearing procedures are critical.
- When descending over 100 feet, power should be brought to idle.
- With power idle, pitch controls airspeed much like with full power.
 - Increase in pitch results in a decrease in airspeed.
 - Decrease in pitch results in an increase in airspeed.

Maneuvering During Slow Flight

- First completion of the maneuver will be a demonstration.
 - Instructor does, Instructor says.
 - Verify correct procedure.
 - Demonstrate turns, climbs, and descents.
 - Points of Emphasis.
 - Pre-maneuver check.

- Selecting visual reference
- Pitch for airspeed, Power for altitude
- Bank angle

Symptoms of a Stall (Incipient and Full)

- Incipient
 - As the airplane approaches the stall, point out the indications of the approaching stall.
 - Decreasing airspeed.
 - Higher than normal angle of attack.
 - Reduced control effectiveness.
 - Stall warning.
 - Light buffeting.
- Full
 - As the airplane enters the full stall stage, point out the indications.
 - Heavy buffeting.
 - Nose drop.
 - High rate of descent.
 - If the airplane is not coordinated → wing drop.

Recovery from a Stall (Incipient and Full)

- For the third stall, recover from the first indication.
- Note the altitude loss
- Compare the altitude loss difference between the three different recoveries.
 - Emphasize to the student that this is why it is important to be aware of early indications of a stall.

Recovery from a Stall (With and Without Power)

- Without Power
 - For the first stall, bring the airplane to a full stall and recover without using power.
 - Note the altitude loss.
- With Power
 - For the second stall, bring the airplane to a full stall and recover using power.
 - Note the altitude loss.

Power-Off Stall Demonstration

- First completion of the maneuver will be a demonstration.
 - Instructor does, Instructor says.
 - Verify correct procedure.
 - Points of Emphasis.
 - Pre-maneuver check.
 - Selecting visual reference.
 - Safe altitude.
 - Ensure proper coordination.

- Explain how improper recovery can lead to a secondary stall.

Power-On Stall Demonstration

- First completion of the maneuver will be a demonstration.
 - Instructor does, Instructor says.
 - Verify correct procedure.
 - Points of Emphasis.
 - Pre-maneuver check.
 - Selecting visual reference.
 - Safe altitude.
 - Ensure proper coordination.

Cross-Controlled Stall Demonstration

- Determine student level of comfortability with maneuvers thus far.
 - If the student is not feeling well, it is not advised to demonstrate a cross-controlled stall.
- Tell the student that this stall is for demonstration purposes and that they should not attempt to replicate it.

Spiral Dive Recovery

- Instructor should put the airplane into a spiraling situation and instruct the student to recover.
 - Recovery:
 - Power idle.
 - Wings level.
 - Pitch up.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 114.
- Assign homework:
 - Read chapter 5 of the Airplane Flying Handbook - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Review the Warrior Maneuvers Manual:

Completion Standards

This lesson is complete when the applicant can:

1. Demonstrate proper sequential checklist procedures for normal procedures without instructor guidance.
2. Correctly perform previously learned maneuvers with minimal instructor guidance.
3. Demonstrate climbing and descending turns with instructor guidance.
4. Maneuver during slow flight with instructor guidance.
5. Recognize the difference between incipient and fully developed stalls and recover from incipient and fully developed stalls with instructor guidance.